

INTELLIGENT EXPERIMENTAL DATA MANAGEMENT SYSTEM WITH CONCURRENT PROCESSING AND ANOMALY DETECTION

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ABSTRACT

This paper presents an Intelligent Experimental Data Management System designed to manage, process, and analyse experimental datasets efficiently. The system integrates object-oriented programming, concurrency techniques, and statistical anomaly detection using Z-score analysis. It enables scalable processing through multiprocessing, threading, and asynchronous execution while providing automated detection of abnormal data values. The proposed system improves traditional data analysis tools by introducing automation, performance optimization, and intelligent statistical reasoning.

1. INTRODUCTION

Experimental data analysis plays a critical role in scientific research and industrial applications. However, traditional tools such as spreadsheets and basic scripts lack scalability, automation, and intelligent data interpretation capabilities.

This project addresses these limitations by developing a structured Experimental Data Management System that integrates software engineering principles with statistical analysis and concurrent computing techniques.

2. RELATED WORK / PROBLEM CONTEXT

Existing data analysis tools:

Lack object-oriented structure

Do not support concurrency

Require manual anomaly detection

Are inefficient for large datasets

This creates a need for a system that is both scalable and intelligent.

3. METHODOLOGY

The system was developed using the following approaches:

3.1. Object-Oriented Programming

Core entities:

Experiment

Trial

Measurement

3.2.Design Patterns

Strategy Pattern for validation

Observer Pattern for logging

3.3. Concurrency Models

Threading for task queue processing

Multiprocessing for statistical computation

Async programming for file operations

3.4. Statistical Analysis

Mean and variance computation

Z-score anomaly detection

4. SYSTEM IMPLEMENTATION

The system was implemented in Python using:

Pandas for data analysis

Multiprocessing for parallel computation

Threading for task management

AsyncIO for non-blocking file operations

The architecture follows a layered structure:

Input Layer

Object Model Layer

Processing Layer

Concurrency Layer

Intelligence Layer

5. RESULTS AND DISCUSSION

The system was tested using a wine quality dataset.

Key results:

Parallel processing reduced computation time compared to sequential execution

Anomaly detection successfully identified abnormal values using Z-score

The system efficiently handled large datasets using generator-based iteration

6. CONTRIBUTION / INNOVATION

The main contribution of this work is the integration of:

Concurrent processing architecture

Statistical computation framework

Automated anomaly detection system

This combination creates an intelligent data management system capable of supporting scientific analysis.

7. CONCLUSION

This paper presents an EDMS that integrates software engineering principles with statistical analysis and concurrency. The system demonstrates improved performance, scalability, and intelligent data interpretation capabilities.

Future work includes integrating machine learning models for predictive analytics and database integration for persistent storage.

REFERENCES

- [Python_3_Object-Oriented_Programming_\(en\)\(1\).pdf](#)